Unit: Unit 6: Systems of Linear Equations and Inequalities **Name:** ____________
Topic: Applications of Systems (Word Problems) **Date:** __________

Foundation (Jalapeno)

Open Ended Questions (FRQ)

Q9. Solve the system of equations: $2x + 3y = 7$ and $x - 2y = -3$. Provide the solution as an ordered pair (x, y).

Q10. A company produces two products: A and B. Each unit of product A requires 2 hours of labor and 3 units of material. Each unit of product B requires 3 hours of labor and 2 units of material. The company has 240 hours of labor and 210 units of material available per week. Write a system of inequalities to represent the number of units of product A (a) and product B (b) that the company can produce. Provide the system of inequalities and explain what each inequality represents.

Chef's Tips

- Make sure students show their work and explain their reasoning
- Emphasize the importance of interpreting solutions in context
- Encourage students to evaluate the reasonableness of their solutions